Delivery 3

Task 1:

a) født → alder
b) født, personnr → fødselsnr
c) stilling, fareT, hemmeligT, skummeltT → lønn
d) stilling, fareT → skummeltT

Task 2:

a) $A^{+} = A, L$
b) $NKA^{+} = N, K, A, L, F, G, P, B$
c) NKA and NAFG

Task 3:

a) Normalform is a classification of how well you minimize the amount of data repeated in your database. The less data is repeated the faster queries become, but the more complicated those queries are (often) since you are then required to utilize joins between tables. There are 4 normalform classifications in ascending order (worst to best): NF1, NF2, NF3, and BCNF.

If rows in your tables repeat or there is an array in a row's cell, then the table is not in NF1. To be in NF2 the structure must satisfy NF1, and then there cannot be a non-key attribute that depends only on a part of a candidate key. To be in NF3 first satisfy NF2 and all non-key attributes dependent only on candidate keys. For BCNF first satisfy NF3, next restriction is that all the attributes must depend on only one candidate key, and not a part of it or whatever.

b) First, we split functional dependencies:

1. agentID → navn	2. agentID → født
3. navn, født → agentID	4. navn → initialer
5. oppdragsNavn → varighet	6. oppdragsNavn → lokasjon

Since it is {agentID, oppdragsNavn} or {navn, født, oppdragsNavn} that are the only candidate keys, and due to number 5 – oppdragsNavn is not a superkey all by itself, the table is not in BCNF. Since due to number 5 – varighet is not a key at all, the table is not in 3NF either. Since due to number 5 – oppdragsNavn is a part of both of candidate key alternatives, the table is not in NF2 either. Thus, we conclude NF1.

Task 4:

First, I make sure that A → BC and BC → A are preserved, so I create $R_1(A, B, C)$. Next, I split D → E and E → F into their own tables $R_2(D, E)$ and $R_3(E, F)$ respectively. Lastly, I join together A and D to indirectly get back the AD → F through $R_4(A, D)$ table.